

MARK SERRANO

Full Stack Developer | Python Developer | AI Application Developer |

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SUMMARY

Full Stack Developer and Software Engineer with 3+ years of experience in software development, cloud technologies, database management and Customer support. Skilled in Python, React.js, Node.js, AWS, MongoDB, and Snowflake, with experience building AI-powered applications, automation solutions, and scalable web systems.

PROFESSIONAL EXPERIENCE

Accenture

Senior AI Data Analyst/Database Administrator

July 2025 – January 2026

- Provided production support and executed database change requests in Snowflake, ensuring accurate and controlled implementation of SQL scripts.
- Monitored and troubleshooted Snowflake and DynamoDB database issues, collaborating with technical teams to maintain database reliability and performance.

WeServ(Fujitsu Philippines)

Application System Engineer

March 2022 – April 2022

- Developed and maintained a SharePoint portal for Learning & Development, improving access to training resources and internal information.
- Automated Project Management (PM) reporting processes, reducing manual reporting effort and improving operational efficiency.
- Collaborated directly with Japanese clients and stakeholders to gather requirements, provide project updates, and coordinate system enhancements.

Fujitsu Limited (Japan)

Software Engineer

July 2023 – August 2024

- Designed and developed an automated file transfer solution using Shell Scripting, running on Solaris OS for a banking client.
- Developed and tested file transfer scripts in a simulated Solaris environment using Oracle VirtualBox, ensuring reliable file processing and transfer operations.
- Collaborated with the Japanese development team to analyze requirements, troubleshoot issues, and deliver the file transfer solution.

PROJECTS

MAK-AI v2 – AI Assistant Platform -Personal Project

Tools: React.js, Python (FastAPI), LangGraph, PostgreSQL, n8n, Groq API, RAG, Git

- Developed a full-stack AI assistant platform using React.js and FastAPI with a modular, scalable architecture.
- Built LangGraph-powered AI agents with conversation memory, state management, and multi-step reasoning capabilities.
- Implemented Retrieval-Augmented Generation (RAG) using vector embeddings and document retrieval for context-aware responses.
- Developed AI-powered workflow generation that creates n8n automation workflows from natural language prompts, enabling users to automate business processes without manually building workflows.
- Integrated AI agents with workflow orchestration, allowing Mak-AI to generate, modify, and execute automation pipelines for common business tasks.
- Designed a ChatGPT-style interface with workspaces, chat history, document uploads, markdown rendering, and streaming AI responses.
- Developed RESTful APIs for chat, workspaces, conversations, and document processing.
- Used PostgreSQL for persistent storage of users, conversations, workspaces, and chat history.

Vistra Corp

Tools: Snowflake, ServiceNow, Python, SQL, Azure AD, AWS Services, Excel

- Administer and optimize Snowflake and DynamoDB databases to ensure performance, security, and scalability.
- Automate integration between Snowflake and ServiceNow CMDB, enhancing data accuracy and operational efficiency.
- Collaborate with clients and stakeholders to understand data requirements and deliver effective database solutions.

IDEA File Transfer System (for Banking – Japan)

Tools: Shell Script, Oracle VirtualBox, Visual Studio Code

- Developed an automated file transfer system for a banking client using shell scripting on Solaris OS.
- Simulated system operations in a virtual environment to ensure reliability pre-deployment.

Automated PM Report System

Tools: PowerApps, Power Automate, SharePoint, HTML, CSS

- Created an automated reminder tool to streamline task management for project managers.
- Integrated with SharePoint to reduce manual follow-ups and boost team accountability.

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- AWS Certified Solutions Architect - Associate

EDUCATION

Bachelor of Science in Information Technology (2016-2020)

Laguna State Polytechnic University, Philippines

TECHNICAL SKILLS

- Frontend:** React.js, JavaScript, HTML5, CSS3, Tailwind CSS, Figma
- Backend & Automation:** Python, FastAPI, Node.js, Express.js, REST APIs, LangGraph, Shell Scripting, JWT Authentication, n8n
- AI & LLM:** LangGraph, AI Agents, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Embeddings, OpenAI API, LLM Integration, Conversation Memory
- Database:** MongoDB, SQL, Snowflake, DynamoDB, PostgreSQL
- Cloud:** AWS (S3, EC2, Lambda, CloudFront, Elastic Beanstalk, Route 53, IAM, DynamoDB, API Gateway, ACM, VPC, CodeBuild, CodePipeline), GitHub Actions
- Tools:** Postman, Visual Studio Code, Oracle VirtualBox, Linux, ServiceNow, Git
- Microsoft Power Platform:** PowerApps, Power Automate, SharePoint